

[illegible]

| No. | Task /Activity                                                                                        | Hazard and Consequence                                                                                                      |                                                               | Initial Risk |                      |             | Mitigating Controls                                                                                                                                                                                                                                                                                                                               | Residual Risk |                      |             |
|-----|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------|----------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------|-------------|
|     | Sequential Steps                                                                                      | Hazard                                                                                                                      | Consequence                                                   | Probability  | Consequence Severity | Risk Rating | Measures to Reduce Risk to ALARP                                                                                                                                                                                                                                                                                                                  | Probability   | Consequence Severity | Risk Rating |
| 1   | Review TRA and TBT                                                                                    | Miscommunication                                                                                                            | Personnel Injury / Environmental Pollution                    | 1            | 2                    | 2           | Situational awareness and proper use of PPE. Hard hats, impact gloves, cotton gloves, latex gloves, steel toed boots, safety glasses, side shields and face shield                                                                                                                                                                                | 1             | 1                    | 1           |
| 2   | Prior to any task being undertaken ensure that all personnel are aware of the hazards involved.       | Electrical shock, possible slips trips and falls. Pinch points poor body position. Hydraulic discharge and spills.          | Personnel Injury / Environmental Pollution                    | 2            | 3                    | 6           | All personnel must be aware of their surrounds and weather conditions. Use proper Manual Handling techniques, i.e., Always use proper body positioning, lift with legs, and obtain second personnel when loads are over 40lbs. Read all SDS on chemicals used, use PPE that is specified in SDS and be aware of where all spill kits are located. | 2             | 2                    | 4           |
| 3   | <i>If task requires working with power tools a JSA is required and or permit to work from vessel.</i> | Personnel injury, pinch points, improper use of tool, damage to equipment, malfunctioning equipment.                        | Personnel Injury / Environmental Pollution / Equipment Damage | 2            | 2                    | 4           | Review <b>DOUS-TRA-017 POWER TOOL USE</b><br><b>Obtain permit to work if required by vessel.</b>                                                                                                                                                                                                                                                  | 1             | 1                    | 1           |
| 4   | Use of hand tools.                                                                                    | Pinch points, cuts, stabs, trips, slips and fall hazards.                                                                   | Personnel Injury / Environmental Pollution / Equipment Damage | 2            | 2                    | 4           | Use proper PPE for the task. Use the proper tool for the task at hand. Keep tools clean and secured when not in use. <b>NEVER</b> leave tools laying around in walkways, staircases or on top of ladders.                                                                                                                                         | 1             | 1                    | 1           |
| 5   | Drain Aux Reservoir to prevent release of fluids to the environment.                                  | Uncontained release of hydraulic fluids slips and falls. High pressure hydraulics, personnel injuries. Damage to equipment. | Personnel Injury / Environmental Pollution / Equipment Damage | 2            | 2                    | 4           | Remove hydraulic oil from Aux reservoir using charge cart in the drain position. Personnel are to keep watch comp as it is draining to prevent damage to comp. Ensure that spill kits and oil absorbent diapers are available and on hand.                                                                                                        | 1             | 1                    | 1           |
| 6   | Remove Hydraulic tooling hoses.                                                                       | Discharge of hydraulic fluid into the environment. Residual pressure in hydraulic lines.                                    | Personnel Injury / Environmental Pollution / Equipment Damage | 2            | 2                    | 4           | Personnel are to carefully bleed pressure from hydraulic lines. Ensure that spill kits and oil absorbent diapers are available and on hand.                                                                                                                                                                                                       | 1             | 1                    | 1           |

|   |                          |                                                                                                        |                                                               |   |   |   |                                                                                                                                                                       |   |   |   |
|---|--------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|---|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 7 | Remove Hydraulic tooling | Heavy tooling, pinch points, improper lifting techniques, poor body positioning. Damage to equipment   | Personnel Injury / Environmental Pollution / Equipment Damage | 2 | 2 | 4 | Utilize two-person lifting technique if equipment is to have to lift by one person. Use proper body and hand placement.                                               | 1 | 1 | 1 |
| 8 | Fill Aux Reservoir       | Uncontained release of hydraulic fluids slips and falls. High pressure hydraulics, personnel injuries. | Personnel Injury / Environmental Pollution / Equipment Damage | 2 | 2 | 4 | Fill Aux reservoir via small charge cart or APU. Ensure that spill kits and oil absorbent diapers are available and on hand. Ensure that all QD caps are put back on. | 1 | 1 | 1 |
| 9 | Clean up work area       | Trips, slips, falls, and dropped objects.                                                              | Personnel Injury / Environmental Pollution / Equipment Damage | 2 | 2 | 4 | Clean up area and secure all tools, ladders if used, ropes. Make sure area is free of debris to prevent accidents.                                                    | 1 | 1 | 1 |

| TRA Approval                                                                                                                                                                        | Name | Position | Signature | Date |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|-----------|------|
| By signing, the Approver confirms that the TRA has been reviewed and considers the task acceptable to proceed with residual risk meeting the acceptance criteria of DO Risk Matrix. |      |          |           |      |

\*Reference DO-HSE-FRM-140 for DeepOcean Risk Assessment Matrix (RAM) and Guidewords